want to examine the Earth at finer and finer resolution—seeking, if not the beings themselves, at least their artifacts.

You look first with a modest telescope, so the finest detail you can resolve is one or two kilometers across. You can make out no monumental architecture, no strange formations, no unnatural reworking of the landscape, no signs of life. You see a dense atmosphere in motion. The abundant water must evaporate and then rain back down. Ancient impact craters, apparent on the Earth's nearby Moon, are almost wholly absent. There must, then, be a set of processes whereby new land is created and then eroded away in much less time than the age of this world. Running water is implicated. As you look with finer and finer definition you find mountain ranges, river valleys, and many other indications that the planet is geologically active. There are also odd places surrounded by vegetation, but which are themselves denuded of plants. They look like discolored smudges on the landscape.

When you examine the Earth at about 100-meter resolution, everything changes. The planet is revealed to be covered with straight lines, squares, rectangles, circles—sometimes huddling along river banks or nestling on the lower slopes of mountains, sometimes stretching over plains, but rarely in deserts or high mountains, and absolutely never in the oceans. Their regularity, complexity, and distribution would be hard to explain except by life and intelligence, although a deeper understanding of function and purpose might be elusive. Perhaps you would conclude only that the dominant life-forms have a simultaneous passion for territoriality and Euclidean geometry. At this resolution you could not see them, much less know them.

Many of the devegetated smudges are revealed to have an underlying checkerboard geometry. These are the planet's cities. Over much of the landscape, and not just in the cities, there is a profusion of straight lines, squares, rectangles, circles. The dark smudges of the cities are revealed to be highly geometrized, with only a few patches of vegetation—themselves with highly regular boundaries—left intact. There are occasional triangles, and in one city there is even a pentagon.